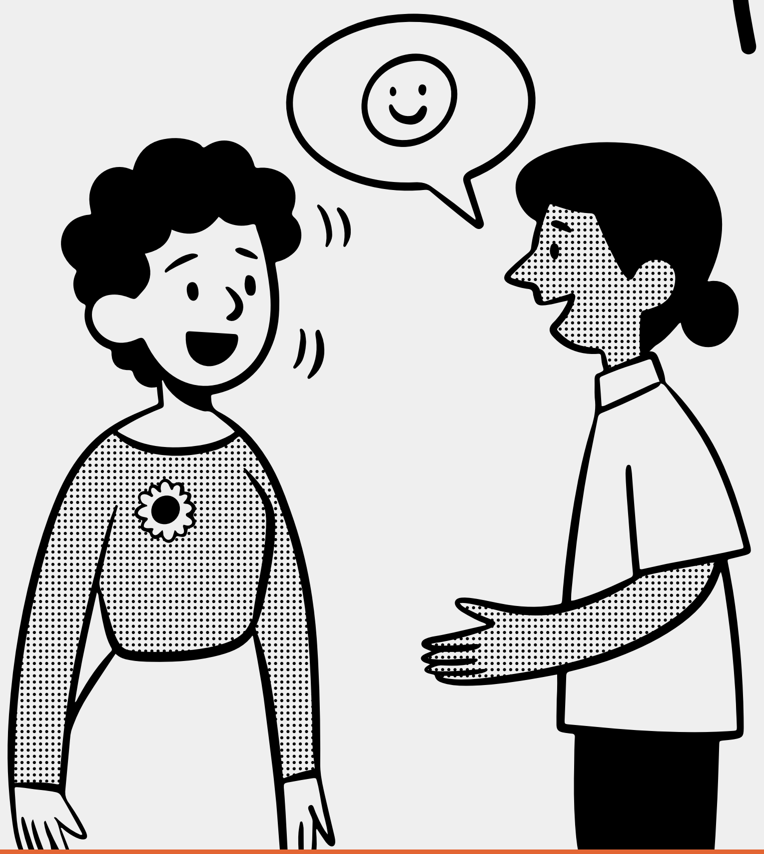
PROJECT 2, SUPPORTING MODEL-BASED DECISION TOOLS SMART SALES TECHNIQUES.



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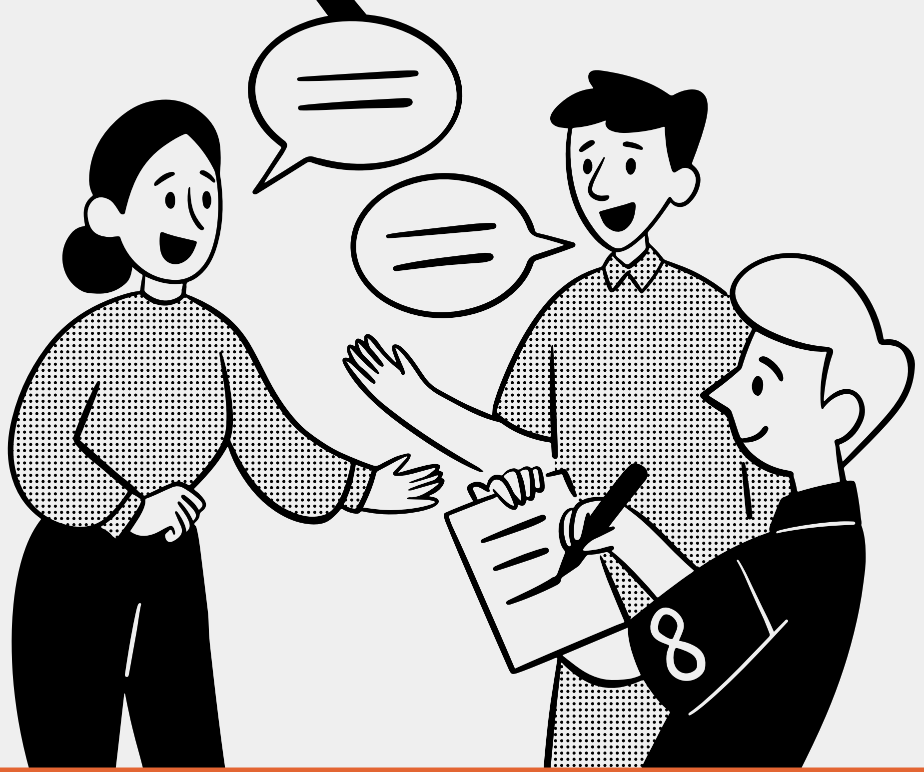

INTRODUCTION

This project addresses Metropolia Smart Transportation's struggles with service unreliability, inefficient resource allocation, and high maintenance costs by building a Decision Support System using Excel and Power Query.

The system will consolidate five disparate operational datasets (routes, vehicles, ridership, maintenance, and feedback) into a unified analytical framework. This integrated data will power interactive dashboards focused on ridership patterns, route efficiency, maintenance drivers, and customer satisfaction. The ultimate goal is to equip the COO with actionable insights, enabling a shift from reactive management to proactive, data-driven decision-making to optimize resources and improve service quality.

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RESEARCH ON THE PUBLIC TRANSPORTATION OPERATION

This research into public transit performance management aims to inform Metropolia's Decision Support System (DSS) by examining established industry best practices. A key finding is that leading transit agencies, such as the Chicago Transit Authority (CTA) and North County Transit District (NCTD), leverage standardized operational data from systems like GPS, Automatic Passenger Counters (APC), Automated Fare Collection (AFC), and Computerized Maintenance Management Systems (CMMS). By doing so, they successfully optimize operations and enhance service quality. The core principle derived from this research is the critical importance of shifting the organizational mindset from reactive reporting to a proactive, data-driven decision-making culture.

STANDARD KPIS IN TRANSIT OPERATIONS (THE 5 DIMENSIONS)

Industry standards reveal that transit performance is monitored across five critical dimensions.

- The first, Service Delivery & Reliability, focuses on metrics like On-Time Performance (OTP), which ideally should be between 85-95%, as well as missed trips and service regularity.
- The second dimension, Ridership & Demand, tracks total passenger boardings and the system's Load Factor, with an optimal peak range of 60-80% capacity.
- Third, Operational Efficiency is measured through financial metrics such as Cost per Passenger Mile and the Farebox Recovery Ratio.
- The fourth dimension, Fleet Management, assesses mechanical health using Mean Distance Between Failures (MDBF), vehicle availability, and fleet age.

Finally, the Customer Experience dimension captures qualitative performance through customer satisfaction scores and complaint rates.

BEST PRACTICES: LESSONS FROM INDUSTRY LEADERS

Examining best practices from industry leaders provides actionable lessons. The Chicago Transit Authority (CTA) demonstrates the value of transparency with its public-facing dashboard, which tracks performance against set targets like 85% OTP and allows for analysis at a granular, route-specific level. From the North County Transit District (NCTD), we learn the importance of integrating operational data with financial performance, enabling year-over-year comparisons and analysis by vehicle type (bus vs. rail). Finally, the Federal Transit Administration (FTA), through its National Transit Database (NTD), provides a crucial lesson in data governance, offering standardized metric definitions for concepts like Load Factor and maintenance types, which ensures all analysis is consistent and comparable over time.

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EFFECTIVE DASHBOARD DESIGN PRINCIPLES

Effective dashboard design relies on several key principles. A successful system must feature a Hierarchy of Information, presenting a high-level executive view with simple traffic-light indicators, a more detailed operational view for daily management, and a deep analytical view that allows for root cause investigation. Interactivity and Filtering are also essential, as they empower users to conduct targeted investigations by filtering data by date, route, or vehicle. Most importantly, the dashboard must produce Actionable Insights. The goal is to enable action, not just passive reporting. This is achieved by highlighting critical alerts, identifying performance trends, and providing comparisons, such as best and worst-performing routes, to guide decision-making.

APPLICATION TO METROPOLIA'S PROJECT

This industry research directly validates and informs the design of Metropolia's project. Our proposed four-report structure focusing on Ridership, Route Efficiency, Maintenance, and Customer Feedback—aligns perfectly with the standard performance dimensions used by leading agencies. The research has informed our design by emphasizing the need for standardized KPI calculations, such as those for Load Factor and OTP. It also confirms that our interactive filtering capabilities, particularly at the route level (like CTA) and by vehicle type (like NCTD), are critical for analysis. Furthermore, the findings reinforce the necessity of integrating operational metrics like OTP with customer feedback to get a complete picture. The strategic focus is therefore confirmed: we must move beyond simple reporting to provide genuine decision support that can identify specific problem routes, like R04 and R09, and highlight clear resource reallocation opportunities.

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METHODOLOGY: ETL

Tools Used: Excel → Power Query → PivotTables → Interactive Dashboards

Process:

1. Extract: import data from Transportation.xlsx

2. Transform

- Standardize data types
- Clean blank/invalid rows
- Add metrics: LoadFactor, VehicleAge, Sentiment Score
- Merge sheets (Route, Vehicle relationships)

3. Load: create relational tables

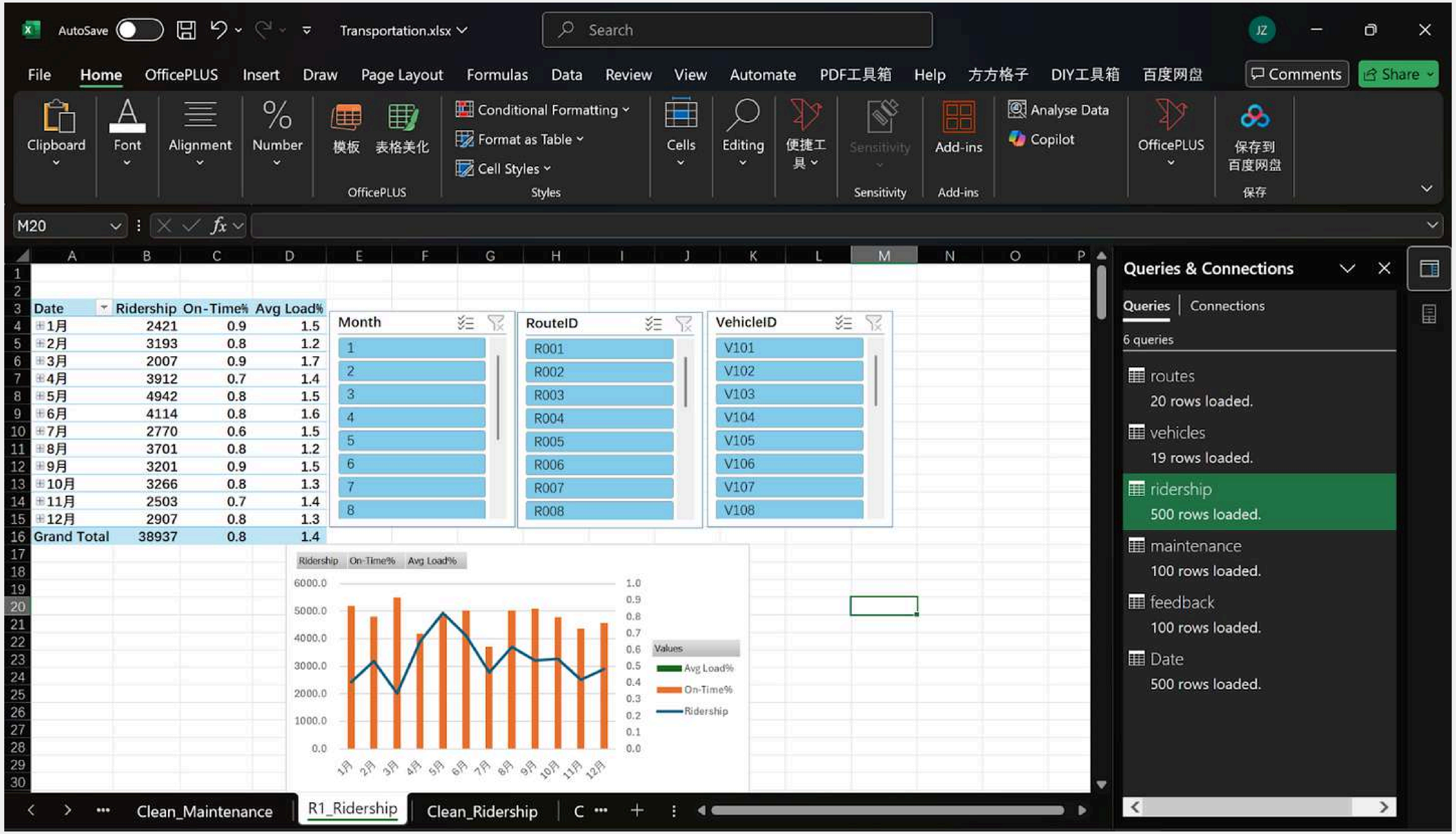
4. Visualize: create reports + unified dashboard

Result: A single trustworthy dataset ready for analysis

DATA VISUALIZATION REPORTS

Report	Focus	Key KPIs
R1	Daily Ridership	On-Time %, Ridership Trend
R2	Route Efficiency	Load Factor, Passenger Demand
R3	Fleet Maintenance	Cost & Work Order Analysis
R4	Customer Feedback	Sentiment Distribution, Avg Rating

R1-DAILY RIDERSHIP



FINDINGS —DAILY RIDERSHIP

Peak demand → OTP decreases sharply

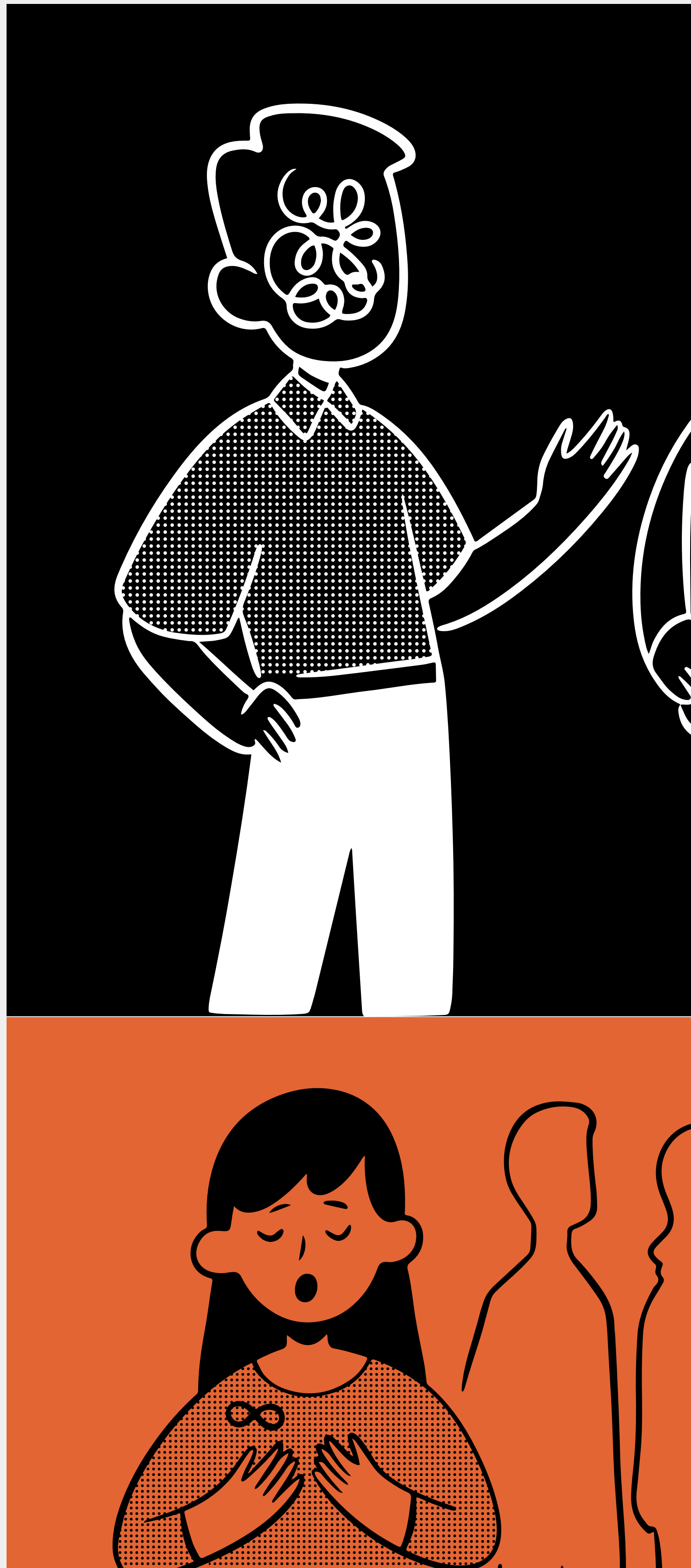
- Weekday morning (07:00–09:00)
- Evening (17:00–19:00)

Critical Routes: R04, R05, R09

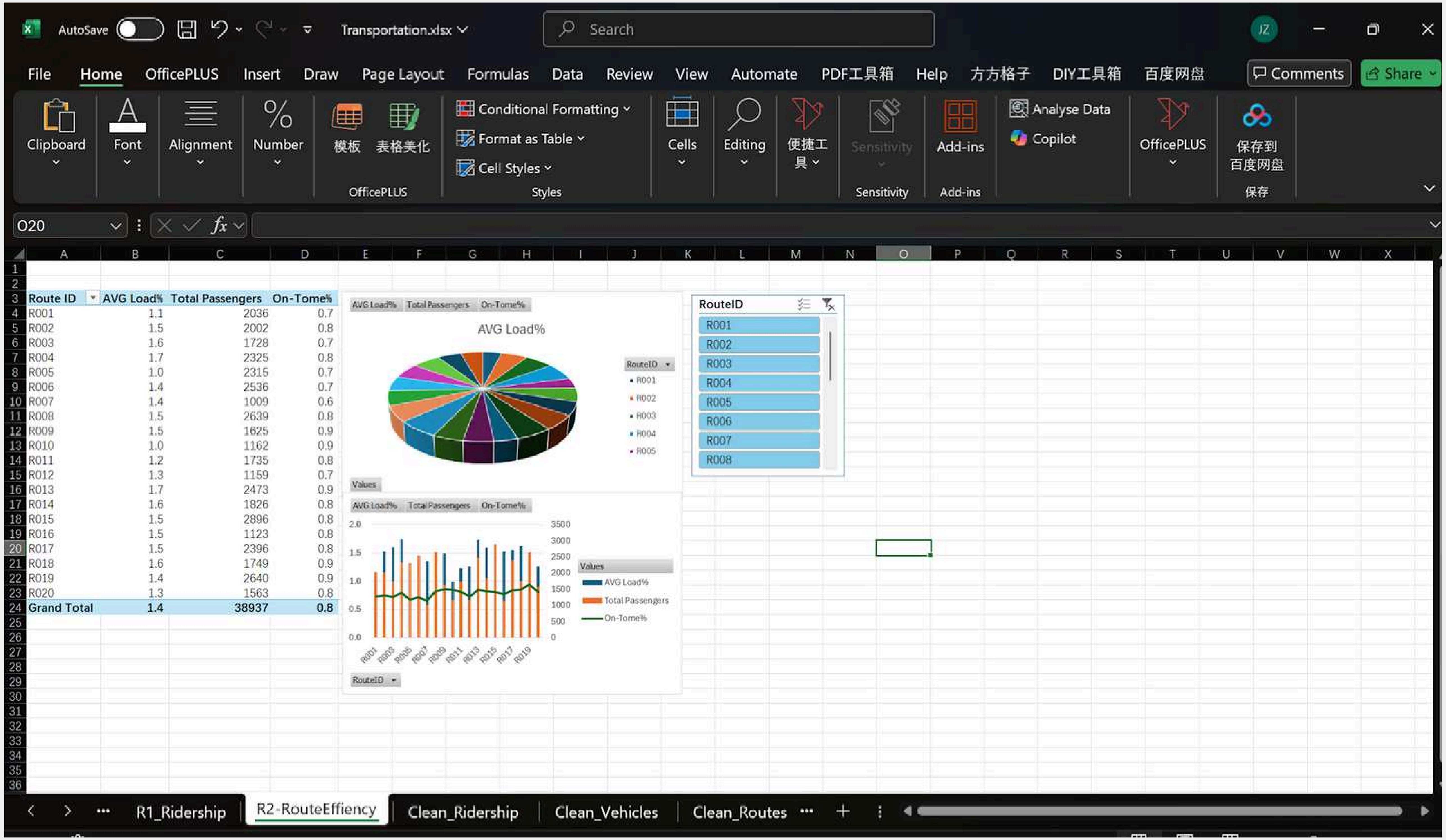
- OTP < 85% benchmark
- Too many passengers → slower boarding → delays

Action Needed:

- Increase fleet during rush hours
- Optimize the schedule to reduce headway variation



R2–CUSTOMER FEEDBACK



Route ID	AVG Load%	Total Passengers	On-Tome%
R001	1.1	2036	0.7
R002	1.5	2002	0.8
R003	1.6	1728	0.7
R004	1.7	2325	0.8
R005	1.0	2315	0.7
R006	1.4	2536	0.7
R007	1.4	1009	0.6
R008	1.5	2639	0.8
R009	1.5	1625	0.9
R010	1.0	1162	0.9
R011	1.2	1735	0.8
R012	1.3	1159	0.7
R013	1.7	2473	0.9
R014	1.6	1826	0.8
R015	1.5	2896	0.8
R016	1.5	1123	0.8
R017	1.5	2396	0.8
R018	1.6	1749	0.9
R019	1.4	2640	0.9
R020	1.3	1563	0.8
Grand Total	1.4	38937	0.8

FINDINGS —CUSTOMER FEEDBACK

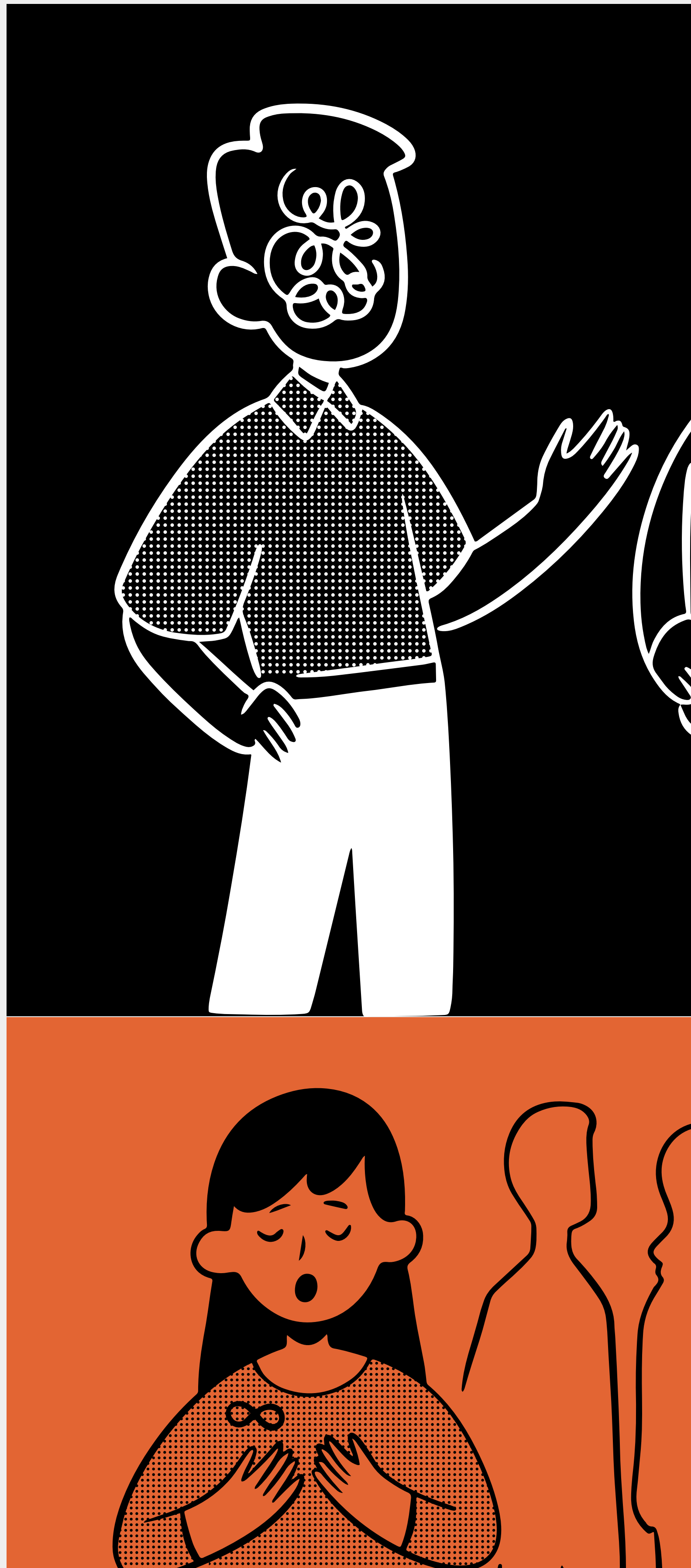
Average Load Factor

- Optimal: 60–80%
- Overcrowded → >90%: R08, R10, R12
- Underutilized → <40%: R02, R06, R16

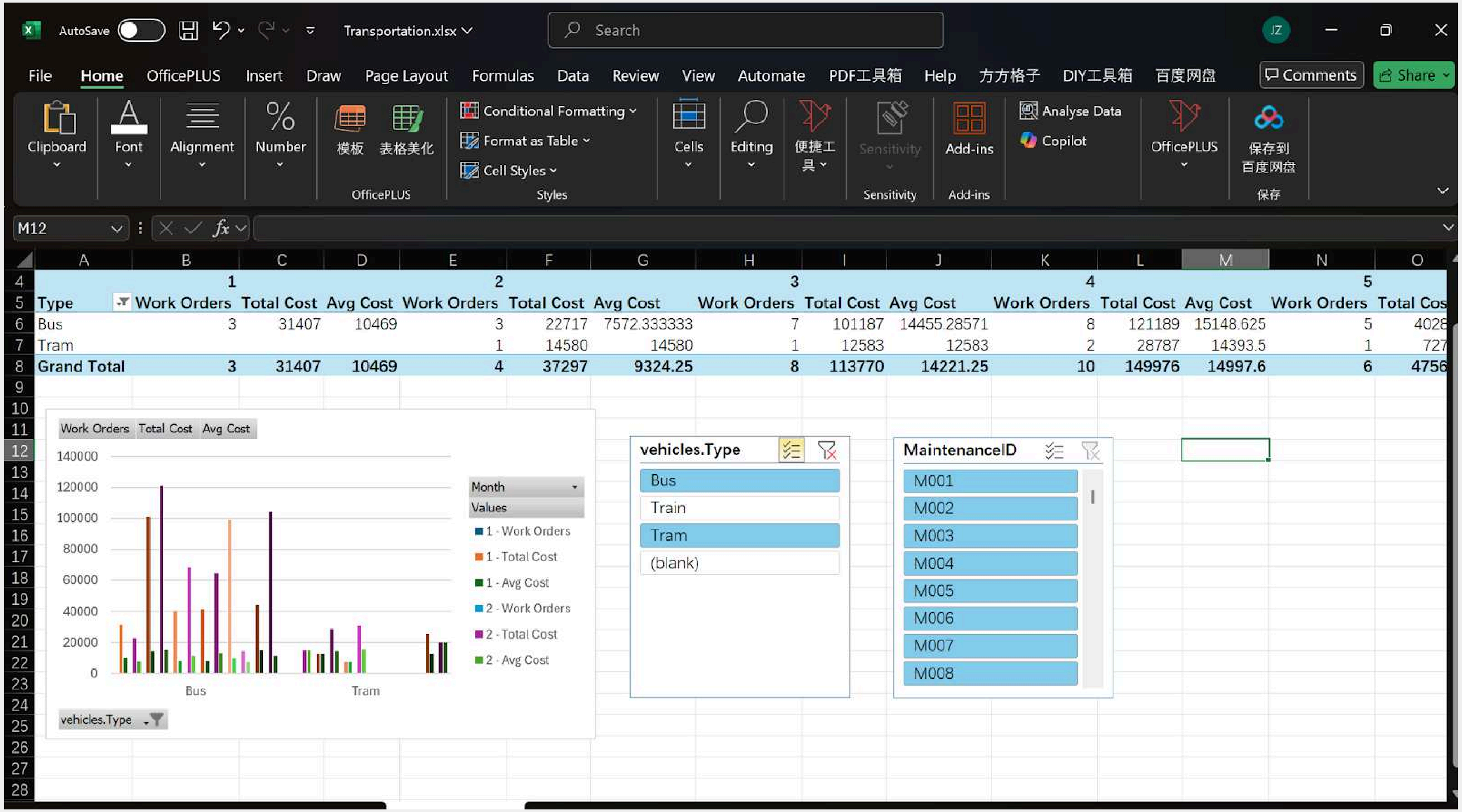
Inference:

- Overcrowded routes need more buses
- Underused routes should reduce frequency or redesign the path

➡ Better ridership balance → higher efficiency & comfort



R3–FLEET MAINTENANCE



FINDINGS —FLEET MAINTENANCE

Maintenance results show:

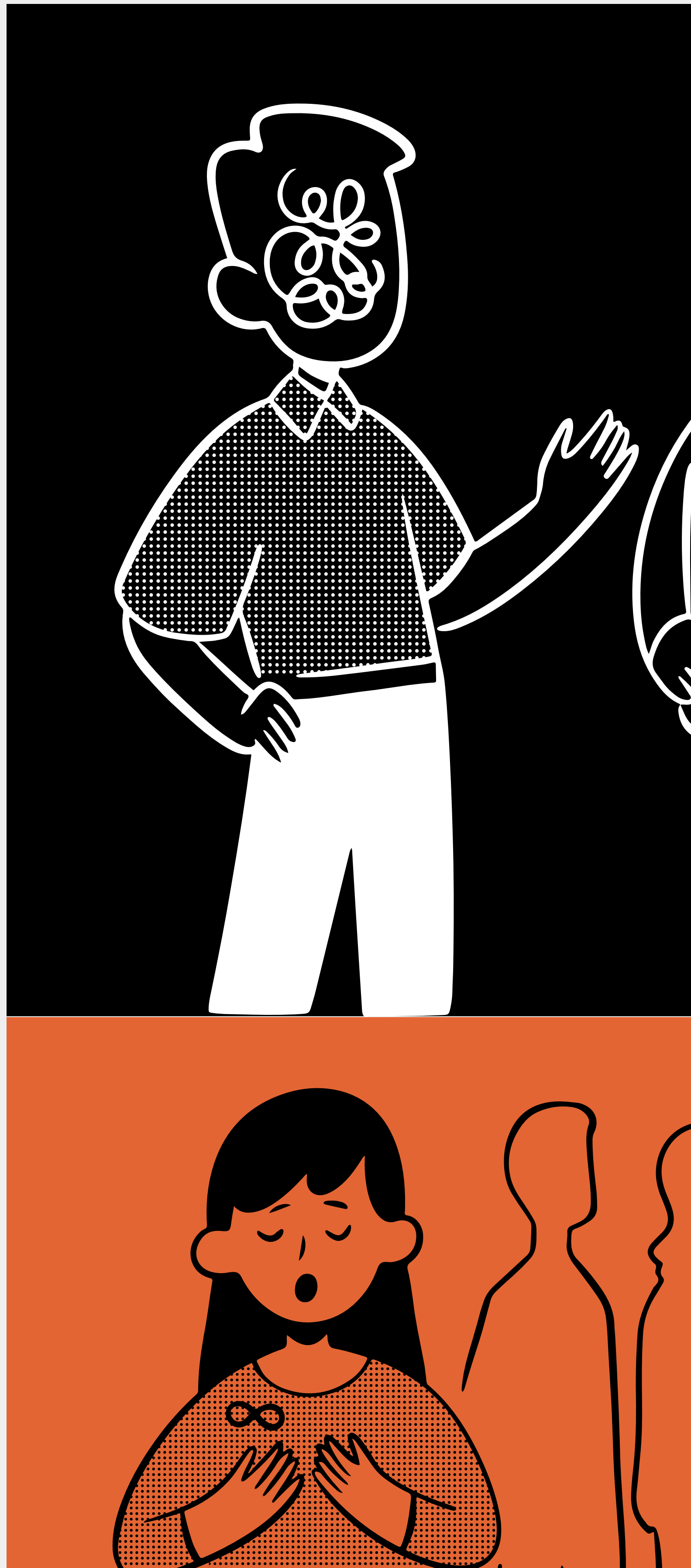
- Buses = Highest Operational Cost
- Trams = Lower work orders, higher reliability

Cost peaks in March & August

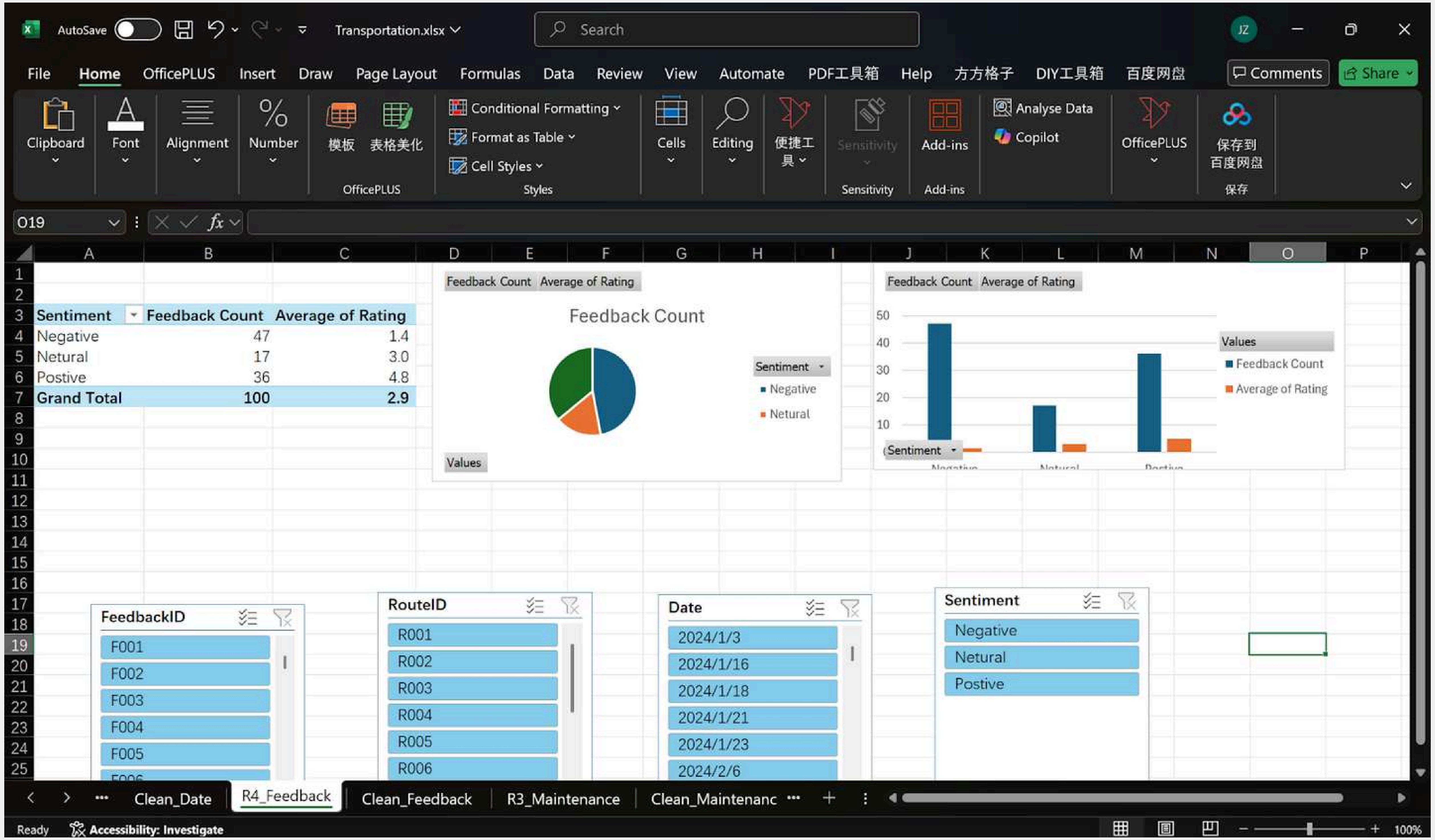
→ **due to seasonal strain and batch servicing**

Recommendations:

- Plan preventive maintenance before peak cost months
- Analyze the cost-to-use ratio to optimize the replacement policy



R4-CUSTOMER FEEDBACK

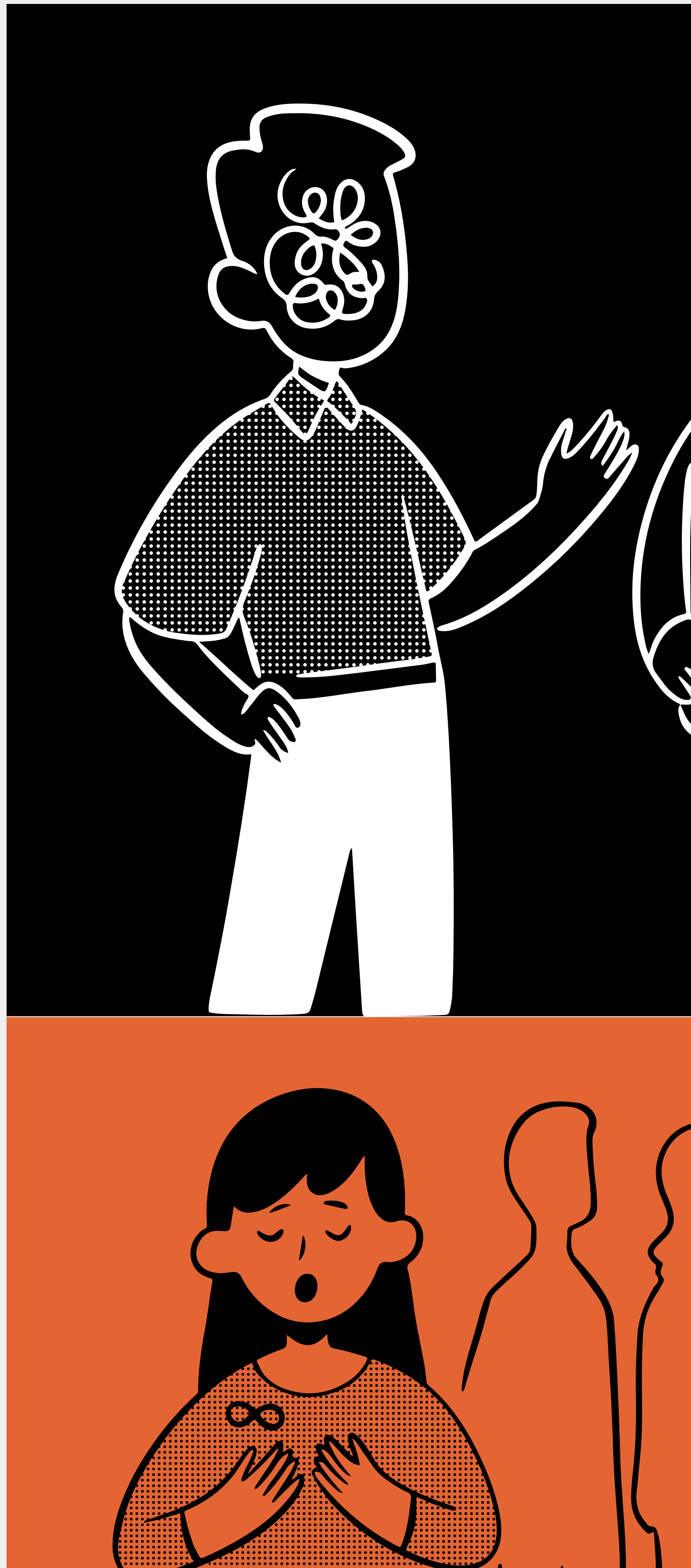


FINDINGS —CUSTOMER FEEDBACK

Sentiment Analysis:

- Positive: 60%
- Neutral: 15%
- Negative: 25% -> complaints about:
  - Late arrival
  - Overcrowding
  - Long wait times

Feedback strongly aligns with: High load factor routes and low OTP performance → Validate dashboard insights from passenger perspective



DASHBOARD DECISION SUPPORT

The unified dashboard allows the COO to:

- Identify root causes instantly
- Monitor KPI thresholds
- Improve operational planning
- Allocate fleet by real demand
- Reduce maintenance cost impact
- Enhance customer experience

DASHBOARD DECISION SUPPORT

RECOMMENDATIONS

Short-term:

- Add vehicles to peak-demand routes
- Improve the headway control system
- Introduce queue-based passenger monitoring

Long-term:

- AI prediction for ridership patterns
- Consider larger vehicles on high-demand routes
- Increase communication channels for passengers

DISCUSSION

The analysis of Metropolia Smart Transportation's data shows how a Decision Support System (DSS) can help the COO make better decisions. By integrating data on routes, vehicles, ridership, maintenance, and customer feedback, the DSS provides a clear and complete view of system performance.

It identifies problem routes with delays (R04, R05, R09), highlights overcrowded (R08, R10, R12) and underused routes (R02, R06, R16) for better resource allocation, monitors fleet maintenance costs with preventive planning, and links customer feedback to operational issues. Overall, the DSS provides actionable insights that support scheduling, resource optimization, cost management, and service quality improvements.

OVERALL INSIGHTS

DSS provides actionable insights for:

- Resource optimization
- Scheduling improvements
- Cost management
- Service quality enhancement

Supports COO in solving operational challenges effectively

CONCLUSION

This project successfully developed a Decision Support System (DSS) for Metropolia Smart Transportation using Excel and Power Query. The system integrates ridership, route efficiency, maintenance, and customer feedback data into an interactive dashboard.

Key Points

- **Integrated Data Analysis:** Combines multiple datasets for a complete operational view
- **Useful Insights:** Highlights busy/underused routes, high-maintenance vehicles, and problem areas
- **Better Decisions:** Links performance data with customer feedback to improve efficiency and satisfaction
- **Scalable Approach:** Can be expanded with more data or predictive analytics

BENEFITS OF DSS

- Supports data-driven decision-making
- Improves service reliability and resource allocation
- Reduces operational costs
- Enhances passenger experience

LESSON LEARNED

- Importance of cleaning and transforming data
- Calculating key performance indicators
- Visualizing data effectively for decision-making
- Understanding how DSS turns complex data into actionable insights

THANK YOU